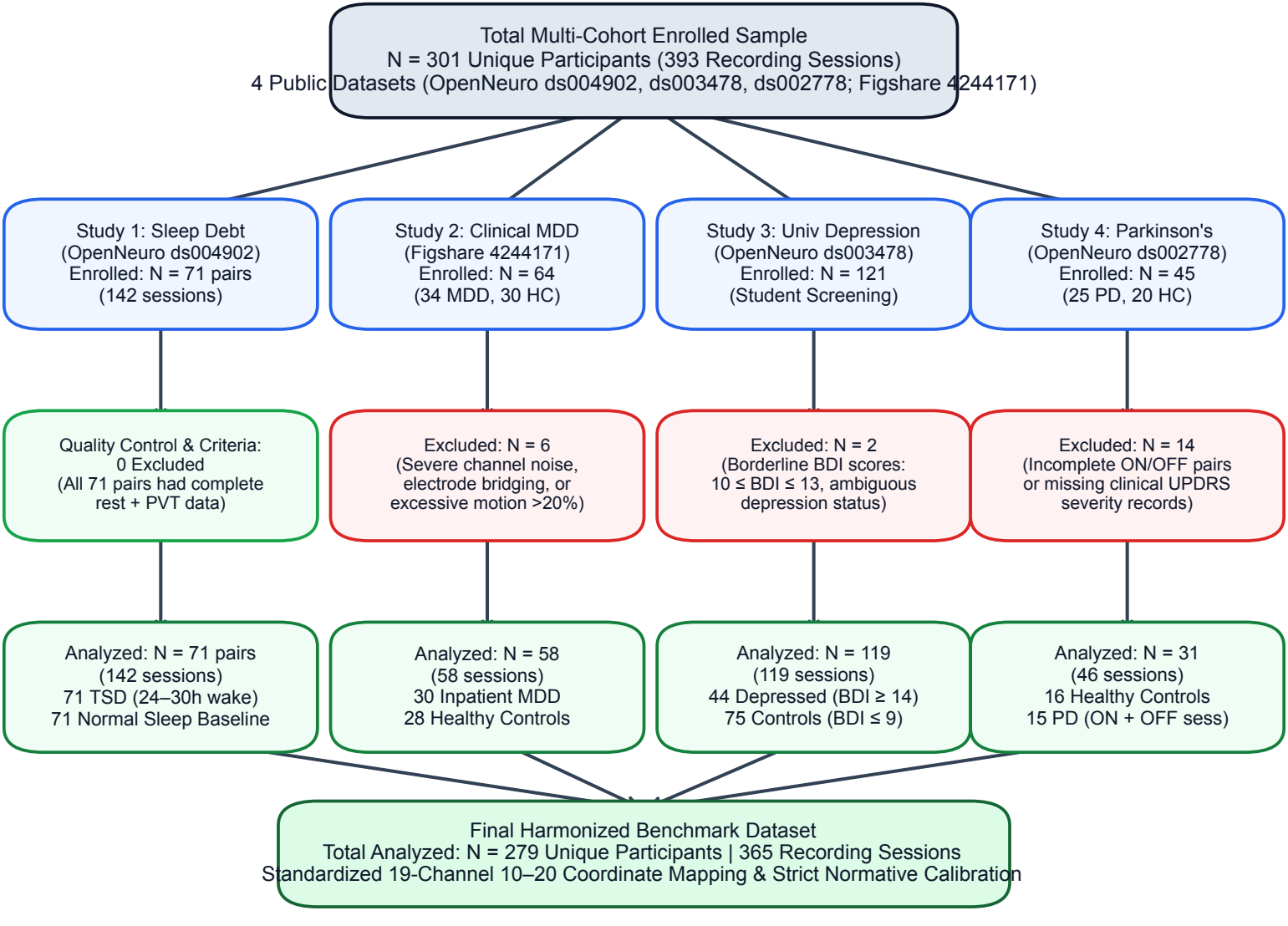
Total Multi-Cohort Enrolled Sample
N = 301 Unique Participants (393 Recording Sessions)
4 Public Datasets (OpenNeuro ds004902, ds003478, ds002778; Figshare 4244171)
Study 1: Sleep Debt
(OpenNeuro ds004902)
Enrolled: N = 71 pairs
(142 sessions)
Study 2: Clinical MDD
(Figshare 4244171)
Enrolled: N = 64
(34 MDD, 30 HC)
Study 3: Univ Depression
(OpenNeuro ds003478)
Enrolled: N = 121
(Student Screening)
Study 4: Parkinson's
(OpenNeuro ds002778)
Enrolled: N = 45
(25 PD, 20 HC)
Quality Control & Criteria:
0 Excluded
(All 71 pairs had complete
rest + PVT data)
Excluded: N = 6
(Severe channel noise,
electrode bridging, or
excessive motion >20%)
Excluded: N = 2
(Borderline BDI scores:
10 ≤ BDI ≤ 13, ambiguous
depression status)
Excluded: N = 14
(Incomplete ON/OFF pairs
or missing clinical UPDRS
severity records)
Analyzed: N = 71 pairs
(142 sessions)
71 TSD (24–30h wake)
71 Normal Sleep Baseline
Analyzed: N = 58
(58 sessions)
30 Inpatient MDD
28 Healthy Controls
Analyzed: N = 119
(119 sessions)
44 Depressed (BDI ≥ 14)
75 Controls (BDI ≤ 9)
Analyzed: N = 31
(46 sessions)
16 Healthy Controls
15 PD (ON + OFF sess)
Final Harmonized Benchmark Dataset
Total Analyzed: N = 279 Unique Participants | 365 Recording Sessions
Standardized 19-Channel 10–20 Coordinate Mapping & Strict Normative Calibration